Call for Proposals (CFP)

Open-H-4D: Collaborative Collection of 4D Cardiac and Respiratory Data

1. Introduction

The Open-H-4D initiative is a new dataset collection effort to accelerate the development of foundational models of dynamic physiology.

Building on the success of Open-H-Embodiment, our goal is to align leading institutions to collect 20,000 total cases of cardiac and/or respiratory motion, including real patient cases and simulated cases. We prioritize 4D CT, MRI, and ultrasound images, while also accepting derived models, simulation motion data, and other relevant data through an open proposal category.

The collected datasets will be made freely available for academic and commercial applications and used to train and validate physics-informed and generative AI models of physiological motion. This data will also be well suited for investigating a diversity of applications, including detecting imaging biomarkers that inform parameters for patient-specific physics, training diagnostic models that incorporate physiological data, and simulating multiple clinical signals such as EKGs and spirometry as well as simulating motion artifacts in 3D images.

NVIDIA will sponsor Open-H-4D by pre-processing all submissions as well as enabling free and open access to the Open-H-4D dataset, foundation model weights, and training code upon project completion.

2. Steering Group

2.1. Leadership

Academic Lead	Alison Marsden, PhD	Stanford
Generative AI Lead	Can Zhao, PhD	NVIDIA
Industry Lead	Mark Palmer, MD, PhD	Ansys, Part of Synopsys
Co-Chair	Mathias Unberath, PhD	Johns Hopkins University
Clinical Lead	Matthew Jolley, MD	Children’s Hospital of Philadelphia
Academic Lead	Paul Segars, PhD	Duke University
Academic Lead	Qi Dou, PhD	The Chinese University of Hong Kong
Co-Chair	Stephen Aylward, PhD	NVIDIA

2.2. Founding Members

Jeff Bodner	Medtronic
Ryan Moore, MD	Cincinnati Children’s Hospital
Xiaoxiao Li, PhD	UBC

3. License

The Open-H-4D dataset and model weights will be publicly available on Hugging Face under the OpenMDW license.

4. What is a foundation model

A **foundation model for physiological motion** is a large AI model pre-trained on diverse 4D data to learn the general principles of how anatomy moves, deforms, and responds to physiological conditions. Instead of being built for one motion-estimation task or one imaging modality, it provides a learned representation of anatomy, dynamics, mechanics, and context that can be adapted to specific patient conditions or to many downstream clinical, simulation, and digital-twin tasks with relatively little task-specific data.

5. Why does data matter for Foundation Models?

Data is central to the success of foundation models because their goal is to learn broad, transferable representations rather than solutions tailored to a specific subpopulation. This requires training on **large and diverse datasets** that extend beyond a single modality, imaging protocol, or patient demographic so the model can capture general structure in the data. With sufficient scale and diversity, the model can learn underlying relationships in the data, allowing it to approximate aspects of the **physical or generative processes** that produce the observations. This enables the model to generalize more effectively across tasks, settings, and downstream applications.

6. Objectives & Outcomes

1. **Data Alignment** – Curate 20,000 total cases, targeting approximately 80% images (16,000 cases), 10% derived models (2,000 cases), and 10% simulated motion data (2,000 cases). These are planning targets; the mix may be adjusted based on the proposals received, including open-category submissions.

2. **Generalist Foundational Models** – Enable training of large, AI models (multi-modal, multi-task) that can be fine-tuned to subpopulations with $\leq$ 10% of current data requirements.

3. **Open-Source Release** – Publish cleaned and pre-processed datasets and model weights under OpenMDW, with the first public release targeted for August 2027.

4. **Community Enablement** - Enable the physiological simulation community, both research and commercial, to create dynamic cardiac and respiratory applications based on foundation models.

7. Scope of Work

Participating teams may propose imaging, derived models, or simulated motion data for either or both of the following task groups:

- Cardiac motion
- Lung respiratory motion

Contributions are accepted in the following categories:

- Images — Preferred contributions are time-resolved 3D CT, MRI, or ultrasound sequences capturing cardiac and/or respiratory motion, with relevant patient metadata.
- Derived models — Image-derived segmentations, anatomical surfaces or meshes, and motion or deformation models are eligible. Derived assets may be proposed when source images cannot be shared, provided the contributor has the rights and approvals required to release the derived assets.
- Simulated motion data — Time-resolved cardiac or respiratory motion from computational models, including simulations based on patient anatomy or virtual subjects, is eligible. Clearly distinguish simulated outputs from measured motion.
- Open category — Other data relevant to modeling physiological motion may be proposed, including paired 3D CT and ultrasound, complementary measurements, other modalities, or other anatomies. Explain the contribution's relevance and how it could support the initiative's models or validation.

For planning, a case represents a patient or simulated subject, with its associated data. Report real patients and simulated cases separately. Multiple time points or image-derived representations of the same patient should not be counted as additional patients; identify related assets and overlapping cases so totals can be reconciled.

8. Proposal and Data Requirements

Each proposal must identify its contribution category or categories and describe the applicable data requirements below. Imaging-specific requirements apply to image submissions; derived, simulated, and open-category proposals should provide the corresponding information for their data.

- Executive Summary
    - High-level description of the data, its source, and contribution category or categories (500 words or less)
    - Table summarizing key quantifications of the data
        - Organ or anatomy: Heart, lung, or other anatomy proposed in the open category
        - Clinical or scientific purpose for acquiring or generating the data
        - Relevant imaging protocol, model derivation method, or simulation approach
        - Number of real patient cases and simulated cases, reported separately
        - Typical number of time points in each 4D sequence
        - Typical spatial and temporal resolution, or corresponding model or simulation resolution
- Data Details
    - Number of real patients, derived-model cases, and simulated cases; identify overlaps between categories and the number of source anatomies used for simulations
        - At least 20 cases must be proposed for case-based datasets. For open-category data not organized by case, specify the amount, unit of contribution, and scientific value for steering-group review.
    - Imaging data
        - Format: DICOM or other
        - Number of time points

        - Coverage of organ of interest: Full or partial
        - Contrast: Yes / No
        - Annotations: Segmentations, landmarks, categorizations
        - Modality specifics
            - CT
                - Low Dose: Yes / No
                - In-plane resolution: 1mm or less
                - Inter-slice spacing:
                    - Cardiac: 2mm or less
                    - Lung: 3mm or less
            - MRI
                - Protocol, Resolution, other
            - Ultrasound
                - Source: 4D TEE, 4D percutaneous, other
    - Derived models
        - Asset type, file format, anatomical coverage, spatial units and coordinates, time points or motion phases, and relationship to any source images that can be shared
        - Source modality and cohort, derivation software and version, segmentation or registration method, quality checks, and known limitations; indicate whether preprocessing or segmentation support is requested
    - Simulated motion data
        - Number of simulated cases and source anatomies, model or solver and version, input anatomy, material properties, boundary conditions, and physiological parameters
        - Output format, spatial and temporal resolution, motion fields or other quantities supplied, validation evidence, and known limitations; identify any patient-derived inputs and permissions
    - Open-category data
        - Data type, quantity and unit of contribution, format, acquisition or generation method, quality checks, and relevance to physiological motion
        - Describe how the data complement images, derived models, or simulations, and identify any requirements above that do not apply
    - Patient information where applicable and available
        - Demographics: Age, gender, weight, height, …
        - Disease state: AFib, PE, cancer, …
        - Vital signs at acquisition: BP, respiratory rate, pulse
        - Relevant treatment history: Post surgery, post radiation therapy, …
        - Reason for scan: Screening, Surgical planning, …
        - For simulated cases, distinguish assigned parameters from measured patient information; mark unavailable or inapplicable fields explicitly.
- Approvals
    - Privacy safeguards for images, metadata, derived assets, and any patient data used to generate simulations
    - IRB/ethics and institutional approval status for acquisition and release; identify pending reviews and their expected completion dates
    - Confirmation that contributed assets can be publicly released for academic and commercial use under OpenMDW. Identify the original data owner and any required third-party permissions, including when a vendor or intermediary supplies the data.

    - Describe remaining approval steps and any restrictions or fees that could affect open release. Deriving models from images does not by itself establish permission to share them.
- Team Qualifications
    - Project lead, contact information, and 4-page-or-less CV
    - Technical support lead, contact information, and 4-page-or-less CV
    - List of relevant team members and their roles
    - Public, free-access links to at most 5 related publications from the team
- Author List and Citation
    - Authors to be listed on the planned publications (see Section 11, Contribution Benefits)
    - Paper to be cited relevant to the data contributed.
- Schedule
    - Delivery method and schedule, including approval milestones, in consideration of the timeline in Section 9

Pre-proposals may be submitted to receive early feedback on data relevance, open-category contributions, and approval or licensing feasibility. Include the applicable Data Details above and identify any unresolved approvals.

Full proposals should be submitted in PDF format. Proposals should not exceed 5 pages, excluding CVs.

Pre-proposals and full proposals must be submitted using this Google Form:
https://forms.gle/Y9pUAHhG7xYCWDoD7

Review criteria include completeness, uniqueness, data quality, size and consistency, suitability for physiological-motion foundation models, and feasibility of obtaining approvals and delivering data on schedule. Image contributions are preferred. The target mix is 80% images, 10% derived models, and 10% simulated motion data; it may be adjusted based on the proposals received. Open-category proposals will be assessed on their scientific value and compatibility with the initiative's release requirements, without a fixed allocation. Within the image category, we will seek an approximate 40% CT, 40% MRI, and 20% ultrasound balance, and an approximate 50/50 balance of lung and cardiac cases across the core collection. These secondary targets may also be adjusted based on the proposals received.

9. Timeline

Milestone	**Date**
CFP released	September 27, 2026
Proposal submission deadline	December 1, 2026
Data collection window	December 1, 2026 – March 1, 2027
Data cleanup and standardization	March 1 – May 1, 2027
Model training and validation	May 1 – July 1, 2027
First public dataset and model release	August, 2027

A public information meeting for prospective proposers is planned approximately one month after MICCAI 2026. The date and joining details will be announced separately.

10. Eligibility and Partnership Models

- Academic institutions, start-ups, and industrial healthcare firms are welcome.
- International participation is encouraged.
- Consortia of multiple organizations may submit a joint proposal with a single lead.

11. Contribution Benefits

Contributing teams will be named co-authors on both the Open-H-4D dataset publication and the follow-up publication describing the resulting Open-H-4D foundation models.

Contributing teams will also receive early-access evaluation checkpoints of the Open-H-4D foundation model, and the dataset itself, for their own downstream experiments.

Contributing teams will receive recognition during the joint dataset and model releases and associated public announcements.

12. Data Governance Ethics and Compliance

- **Patient Privacy:** All human-subject data must be de-identified to HIPAA Safe Harbor standards or equivalent.
- **Regulatory:** Proposers are responsible for obtaining IRB/ethics and institutional approvals where required. Proposals may identify pending approvals, but all required permissions must be in place before data transfer and public release.
- **Licensing:** Final datasets and model weights will be released under OpenMDW (https://openmdw.ai/). Contributors must hold the rights needed to release each contributed asset under that license. This applies to original images, derived models, simulation outputs, and open-category data.

13. Contact Information

- Questions
    - Stephen Aylward: saylward@nvidia.com